Master of Computer Applications

Semester – III

Data Analytics for Business

Experiment No: 5

Title: Walmart Sales Dashboard

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Walmart Sales Dashboard

Project Overview

The Walmart Sales Dashboard is an interactive Business Intelligence solution developed using Microsoft Power BI. The dashboard analyzes Walmart's weekly sales data from 45 stores between 2010 and 2012. It helps understand sales performance, seasonal trends, fuel price variations, CPI, unemployment, holiday sales, and temperature patterns.

The dashboard enables managers and business analysts to monitor key performance indicators (KPIs), compare yearly performance, identify seasonal demand, and make informed business decisions using interactive visualizations and filters.

1. Introduction

The **Walmart Sales Dashboard** is an interactive Power BI dashboard developed to analyze and monitor Walmart's sales performance. The dashboard transforms historical sales data into meaningful visual insights that can support management and sales analysts in understanding sales trends, store performance, economic conditions, and seasonal patterns.

The dashboard provides a single view of important business indicators such as **weekly sales, number of stores, CPI, unemployment, holiday periods, temperature, and fuel prices**. Interactive filters allow users to explore the data for different dates and months.

2. Problem Statement

As a Sales Analyst for a retail chain, the objective is to design an interactive dashboard that enables the sales director to monitor business performance and make data-driven decisions. The dashboard should provide a clear understanding of overall sales, store performance, seasonal trends, and external factors that may influence sales.

Since the Walmart dataset mainly contains store-level sales and economic information rather than product-level profit data, the dashboard focuses primarily on **sales performance, store analysis, time trends, and economic indicators**.

3. Objective

- Analyze overall weekly sales across Walmart stores.
- Compare yearly sales performance.
- Study the impact of Fuel Price, CPI, Temperature, and Unemployment on sales.
- Identify seasonal and holiday sales patterns.
- Build an interactive dashboard for business decision-making.

4. Dataset Description

The Walmart dataset contains historical information related to **45 Walmart stores**. The important attributes used in the dashboard include:

Store: Identifies individual Walmart stores.

Date: Represents the date on which weekly sales were recorded.

Weekly_Sales: Represents the total weekly sales generated by a store.

Holiday_Flag: Indicates whether the particular week was a holiday week.

Temperature: Shows the temperature recorded during the sales period.

Fuel_Price: Represents fuel prices during the corresponding period.

CPI: Represents the Consumer Price Index.

5. Dashboard Design

The dashboard follows a **purple-themed visual design** with KPI cards positioned at the top, filters on the left, and analytical charts in the center and lower sections. This layout allows users to quickly view important performance indicators before moving to more detailed analysis.

The dashboard includes **date and month slicers**, which make it interactive. When a user selects a particular date range or month, the related visualizations are automatically updated.

Data import , Cleaning ,Transformation

Step 1: Import Dataset

- Imported the Walmart.csv dataset into Power BI using Get Data → CSV.
- The dataset contains 6,435 records and 8 columns:
    - Store
    - Date
    - Weekly_Sales
    - Holiday_Flag
    - Temperature
    - Fuel_Price
    - CPI
    - Unemployment

Step 3: Data Cleaning

The following cleaning operations were performed:

- Removed duplicate rows.
- Checked for blank and null values.
- Removed unnecessary spaces from column names.
- Renamed columns for readability where required.
- Verified numerical columns for incorrect values.

Step 4: Data Transformation

Using Power Query:

- Converted Date into Date format.
- Extracted:
  - Year
  - Month
- Sorted months chronologically.
- Created calculated measures using DAX:
  - Total Weekly Sales
  - Average CPI
  - Maximum Fuel Price
  - Count of Stores
  - Total Unemployment

RESULTS / BUSINESS INSIGHTS

Key Findings

- Walmart generated approximately 6.74 Billion in total weekly sales during 2010–2012.
- The company operates across 45 stores, demonstrating a broad retail network.
- 2011 recorded the highest contribution to total sales.
- Sales increase significantly during holiday periods, indicating the effectiveness of seasonal promotions.
- Fuel prices showed a continuous increase, which may impact transportation and logistics costs.
- Average CPI remained stable, suggesting limited inflationary effects on sales.
- Temperature changes indicate seasonal customer purchasing behavior.
- Interactive filters allow managers to analyze sales by month and by individual store.

WALMART DASHBOARD

CONCLUSION & LEARNING OUTCOMES

Conclusion

The Walmart Sales Dashboard effectively summarizes key business metrics using interactive Power BI visualizations. The dashboard provides valuable insights into weekly sales, fuel prices, CPI, unemployment, and seasonal trends. KPI cards help monitor important business indicators, while charts simplify trend analysis across months and years. Interactive slicers improve data exploration and decision-making. Overall, the dashboard serves as an efficient business intelligence solution for monitoring Walmart's retail performance.

Learning Outcomes

- Learned to import datasets into Power BI.
- Performed data cleaning and transformation using Power Query Editor.
- Created calculated measures using DAX formulas.
- Designed interactive dashboards using KPI cards, slicers, and multiple chart types.
- Understood how to interpret sales trends and convert raw data into meaningful business insights.
- Improved skills in business analytics and data visualization using Microsoft Power BI.

GITHUBLINK

https://github.com/prajwaldotcom/PowerBi-DashBoards/blob/main/ASSIGNMENT%205.pbix